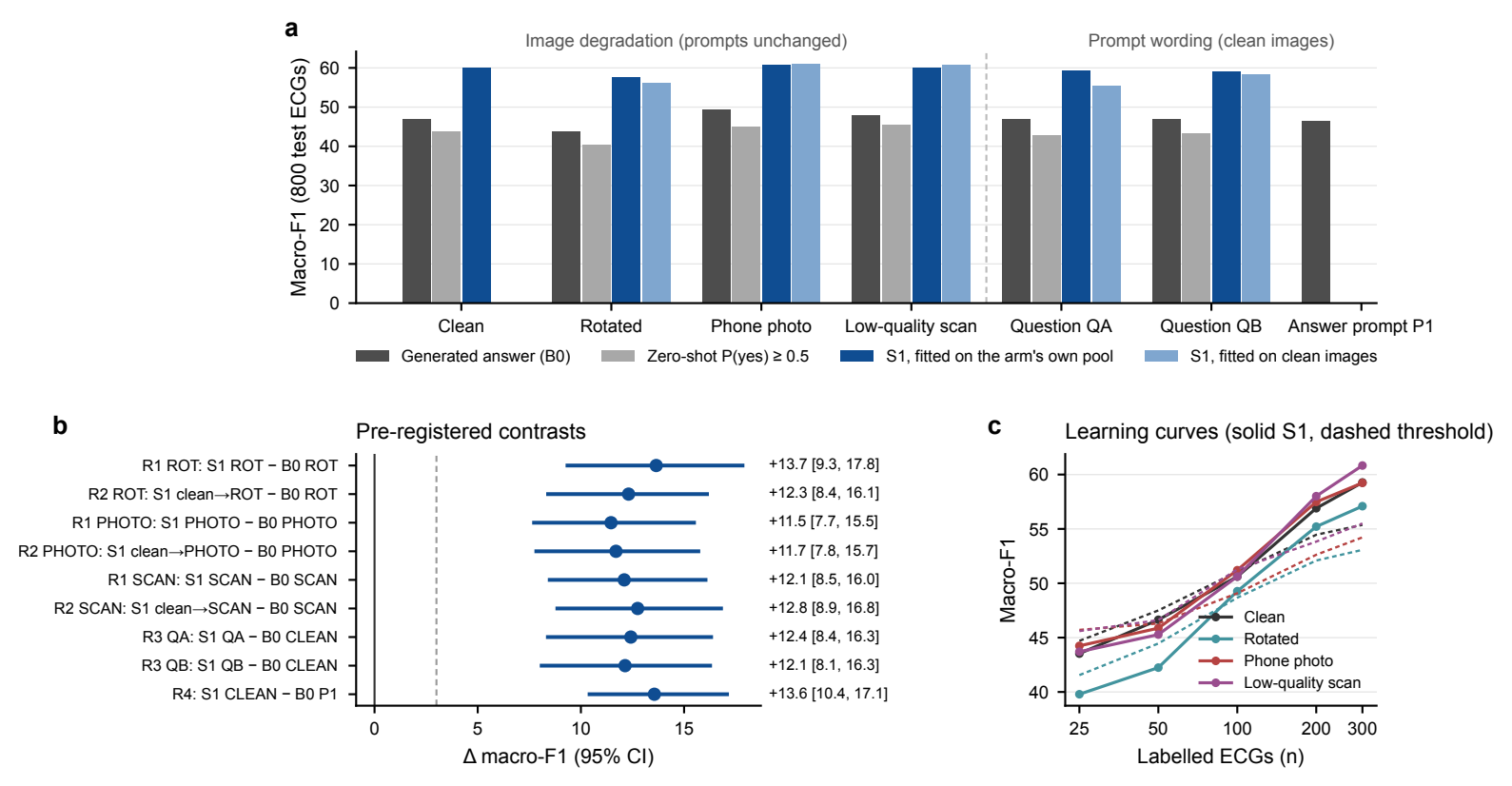
a
Image degradation (prompts unchanged)
Prompt wording (clean images)
Macro-F1 (800 test ECGs)
0
10
20
30
40
50
60
Clean
Rotated
Phone photo
Low-quality scan
Question QA
Question QB
Answer prompt P1
Generated answer (B0)
Zero-shot P(yes) ≥ 0.5
S1, fitted on the arm's own pool
S1, fitted on clean images
b
Pre-registered contrasts
R1 ROT: S1 ROT − B0 ROT
R2 ROT: S1 clean→ROT − B0 ROT
R1 PHOTO: S1 PHOTO − B0 PHOTO
R2 PHOTO: S1 clean→PHOTO − B0 PHOTO
R1 SCAN: S1 SCAN − B0 SCAN
R2 SCAN: S1 clean→SCAN − B0 SCAN
R3 QA: S1 QA − B0 CLEAN
R3 QB: S1 QB − B0 CLEAN
R4: S1 CLEAN − B0 P1
+13.7 [9.3, 17.8]
+12.3 [8.4, 16.1]
+11.5 [7.7, 15.5]
+11.7 [7.8, 15.7]
+12.1 [8.5, 16.0]
+12.8 [8.9, 16.8]
+12.4 [8.4, 16.3]
+12.1 [8.1, 16.3]
+13.6 [10.4, 17.1]
0
5
10
15
Δ macro-F1 (95% CI)
c
Learning curves (solid S1, dashed threshold)
Macro-F1
40
45
50
55
60
25
50
100
200
300
Labelled ECGs (n)
Clean
Rotated
Phone photo
Low-quality scan